

PostgreSQL + pgAdmin4 in Docker (macOS) 🚀

Step 1: Verify Docker is Installed

Open Terminal: `docker --version`

Expected:

Docker version xx.xx.xx

Check Docker is running:

`docker ps`

If no error appears, Docker is running.

Step 2: Download PostgreSQL Image

`docker pull postgres:16`

Verify:

`docker images`

You should see:

postgres 16

Step 3: Create a Docker Network

`docker network create postgres-network`

Verify: `docker network ls`

Step 4: Create a Persistent Volume

`docker volume create postgres-data`

Verify: `docker volume ls`

Step 5: Start PostgreSQL Container

```
docker run -d \  
--name postgres-db \  
--network postgres-network \  
-e POSTGRES_USER=tejasjawale \  
-e POSTGRES_PASSWORD=Tejas@1299 \  
-e POSTGRES_DB=frm \ #Database Name  
-p 5433:5432 \  
-v postgres-data:/var/lib/postgresql/data \  
postgres:16
```

```
docker run -d --name postgres-db -e POSTGRES_USER=tejasjawale -e  
POSTGRES_PASSWORD='Tejas@1299' -e POSTGRES_DB=frm -p 5433:5432 postgres:16
```

Step 6: Verify PostgreSQL

`docker ps`

You should see:

postgres-db

Check logs:

`docker logs postgres-db`

Wait until you see:

database system is ready to accept connections

Step 7: Download pgAdmin4 Image

`docker pull dpage/pgadmin4`

Step 8: Start pgAdmin4 Container

```
docker run -d \  
--name pgadmin4 \  
--network postgres-network \  
-p 5050:80 \  
-e PGADMIN_DEFAULT_EMAIL=tejas.jawale@auisy.com \  
-e PGADMIN_DEFAULT_PASSWORD=Tejas@1299 \  
dpage/pgadmin4
```

```
docker run -d --name pgadmin4 -p 5050:80 -e  
PGADMIN_DEFAULT_EMAIL=tejas.jawale@auisy.com -e  
PGADMIN_DEFAULT_PASSWORD=Tejas@1299 dpage/pgadmin4
```

Step 9: Verify pgAdmin4

`docker ps`

You should see:

postgres-db
pgadmin4

Step 10: Open pgAdmin

Open your browser:

<http://localhost:5050>

Login:

Email :tejas.jawale@auisy.com

Password : Tejas@1299

Step 11: Add PostgreSQL Server in pgAdmin

Click:

Servers → Register → Server

General Tab

Name: [Local PostgreSQL](#)

Connection Tab

Field	Value
Host Name	postgres-db
Port	5433 or 5432
Maintenance DB	frm
Username	tejasjawale
Password	Tejas@1299

Enable:

Save Password = Yes

Click **Save**.

Stop Both Containers

```
docker stop postgres-db pgadmin4
```

No containers should be running.

```
docker ps
```

Start PostgreSQL

```
docker start postgres-db
```

Start pgAdmin

```
docker start pgadmin4
```

start both together:

```
docker start postgres-db pgadmin4
```

Verify They Are Running

```
docker ps
```

Open pgAdmin : <http://localhost:5050>

Restart both containers

```
docker restart postgres-db pgadmin4
```

PostgreSQL and pgAdmin Docker Setup on Mac

This document contains fresh installation steps and login credentials.

Step 1: Remove Existing Containers

```
docker rm -f postgres-db pgadmin4
```

Step 2: Remove Existing Volume (Optional)

```
docker volume rm postgres-data
```

Step 3: Remove Existing Network (Optional)

```
docker network rm postgres-network
```

Step 4: Create Docker Network

```
docker network create postgres-network
```

Step 5: Create Docker Volume

```
docker volume create postgres-data
```

Step 6: Start PostgreSQL Container

```
docker run -d \  
  --name postgres-db \  
  --network postgres-network \  
  -e POSTGRES_USER=tejasjawale \  
  -e POSTGRES_PASSWORD='Tejas@1299' \  
  -e POSTGRES_DB=frm \  
  -p 5433:5432 \  
  -v postgres-data:/var/lib/postgresql/data \  
  postgres:16
```

Step 7: Verify PostgreSQL Logs

`docker logs postgres-db`

Step 8: Start pgAdmin4 Container

```
docker run -d \  
  --name pgadmin4 \  
  --network postgres-network \  
  -p 5050:80 \  
  -e PGADMIN_DEFAULT_EMAIL=tejas.jawale@auisy.com \  
  -e PGADMIN_DEFAULT_PASSWORD='Tejas@1299' \  
  dpagadmin4
```

Step 9: Open pgAdmin

`http://localhost:5050`

pgAdmin Login Credentials

Email: tejas.jawale@auisy.com

Password: Tejas@1299

PostgreSQL Credentials

Database:

Username: tejasjawale

Password: Tejas@1299

Host from pgAdmin: postgres-db

Host from Mac: localhost

Port: 5433

Redis Container Setup

Create Redis container:

```
docker run -d \  
  --name redis-insurance \  
  -p 6379:6379 \  
  redis:7
```

Check container:

`docker ps`

Test Redis:

`redis-cli ping`

Expected:

PONG

Check Insurance cache keys:

```
redis-cli KEYS "insurance:apis:*
```

Start existing Redis container:

```
docker start redis-insurance
```

Stop Redis container:

```
docker stop redis-insurance
```